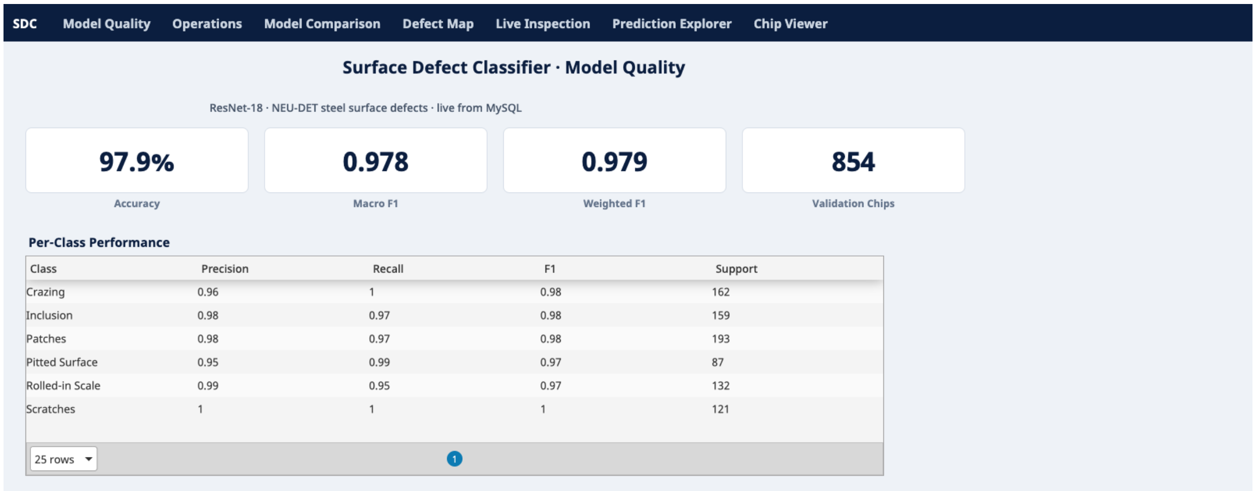
SURFACE DEFECT CLASSIFIER

Chip-level steel surface defect classification with an Ignition Perspective dashboard

Chris Brantner

June 2026

Contents



Class	Precision	Recall	F1	Support
Crazing	0.96	1	0.98	162
Inclusion	0.98	0.97	0.98	159
Patches	0.98	0.97	0.98	193
Pitted Surface	0.95	0.99	0.97	87
Rolled-in Scale	0.99	0.95	0.97	132
Scratches	1	1	1	121

Figure 1: Model Quality

This project classifies steel surface defects from the NEU-DET benchmark at the chip level, then feeds the results into an interactive Ignition Perspective dashboard backed by MySQL. The pipeline parses the Pascal VOC annotations, crops each defect region into a chip, fine-tunes two image classifiers (ResNet-18 and ViT-B/16), and exports the evaluation artifacts. An ETL job loads those into a MySQL star schema, and a seven-view Perspective application reads from it live.

The Dashboard

The dashboard is a seven-view Perspective application that reads from MySQL live. The views cover model quality, plant operations, model comparison, defect localization, a replayed real-time feed, a filterable prediction explorer, and the chips the model got wrong.

Model Quality

Accuracy and F1 KPI tiles, plus a per-class precision/recall/F1 table. All of it reads from the database.

Operations

Per-machine defect counts and cost, a predicted-class Pareto bar chart, and defect volume by day. This is the MES-style view a line supervisor would watch.

Model Comparison

ResNet-18 against ViT-B/16 as grouped horizontal bars for each class, sitting above the live overall-metrics table. The ViT shortfall is easy to see (Pitted Surface drops to 0.57 against ResNet's 0.97).

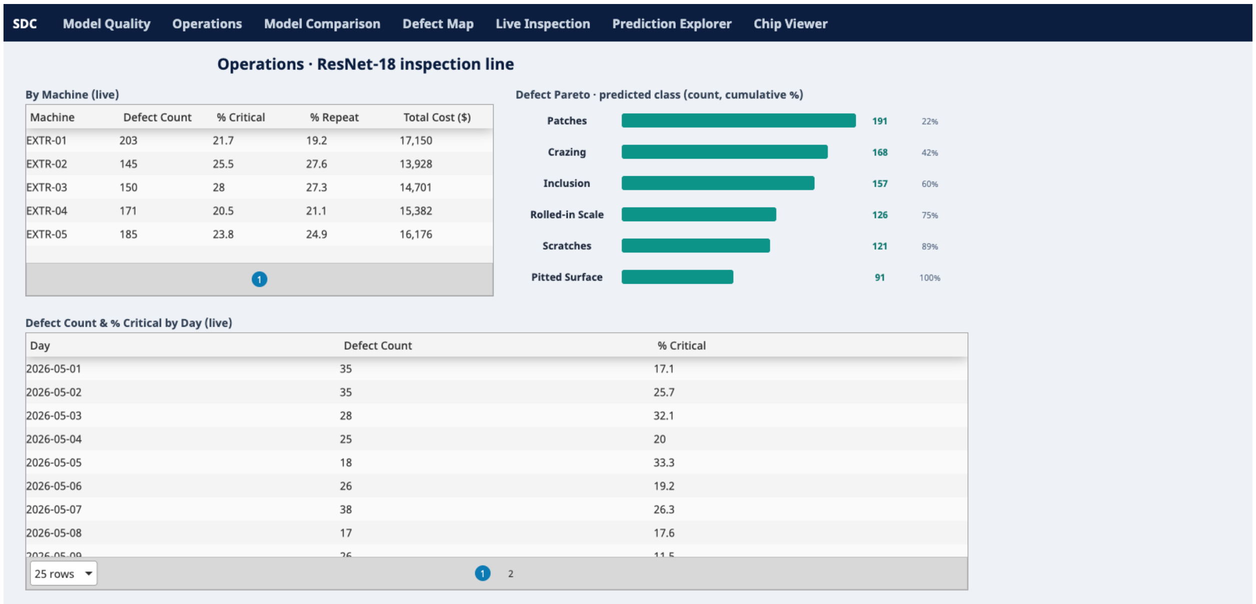


Figure 2: Operations

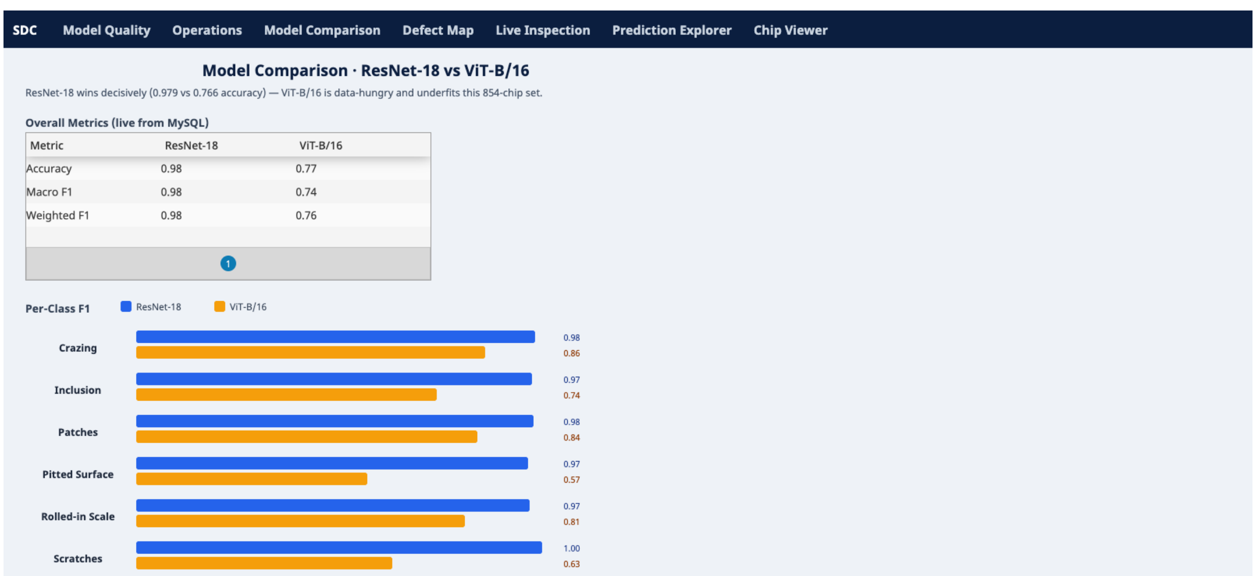


Figure 3: Model Comparison

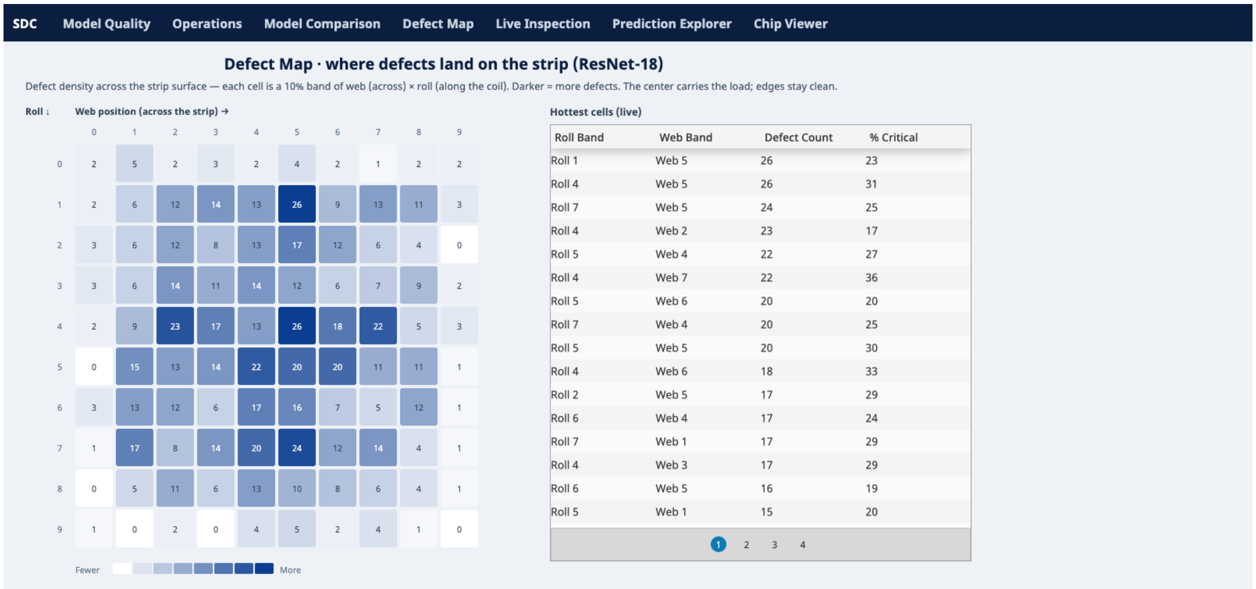


Figure 4: Defect Map

Defect Map

A 10x10 density heatmap of where defects land on the strip. Each defect's bounding-box center is binned by web (across) and roll (along the coil). The center carries most of the defects and the edges stay clean.

Live Inspection

A replayed real-time defect feed (polls every 2s) with running per-class counts and a critical rate.

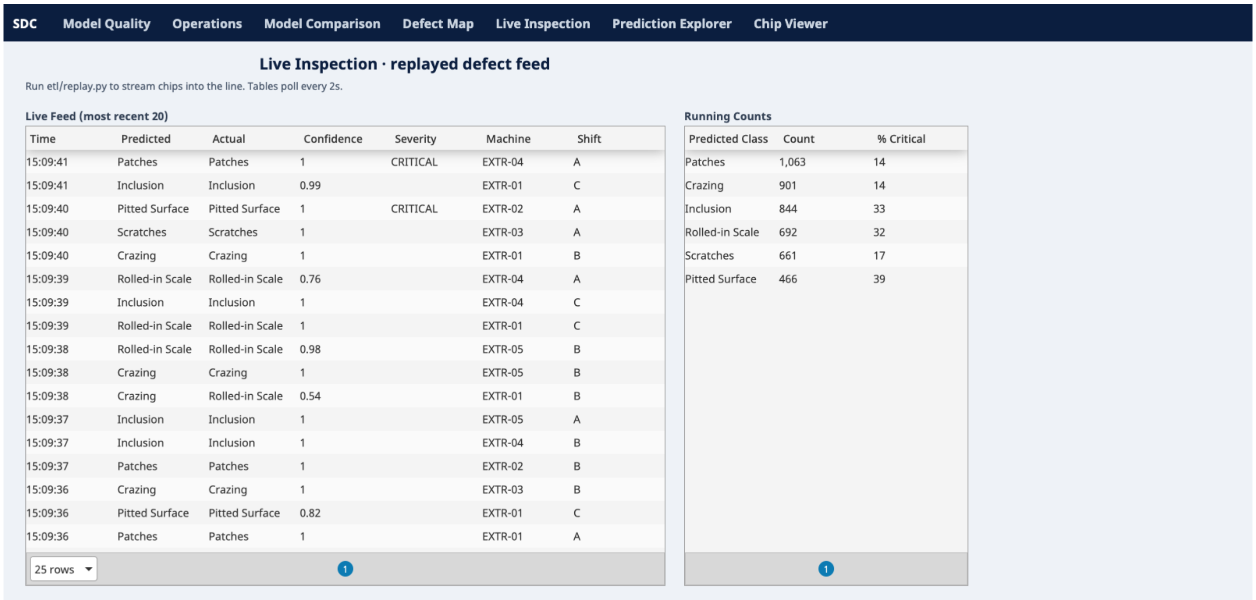
Prediction Explorer

All 854 validation predictions, sortable and filterable.

Chip Viewer

The actual chips ResNet-18 got wrong, ordered by confidence. These are the cases I'd bring to a model review.

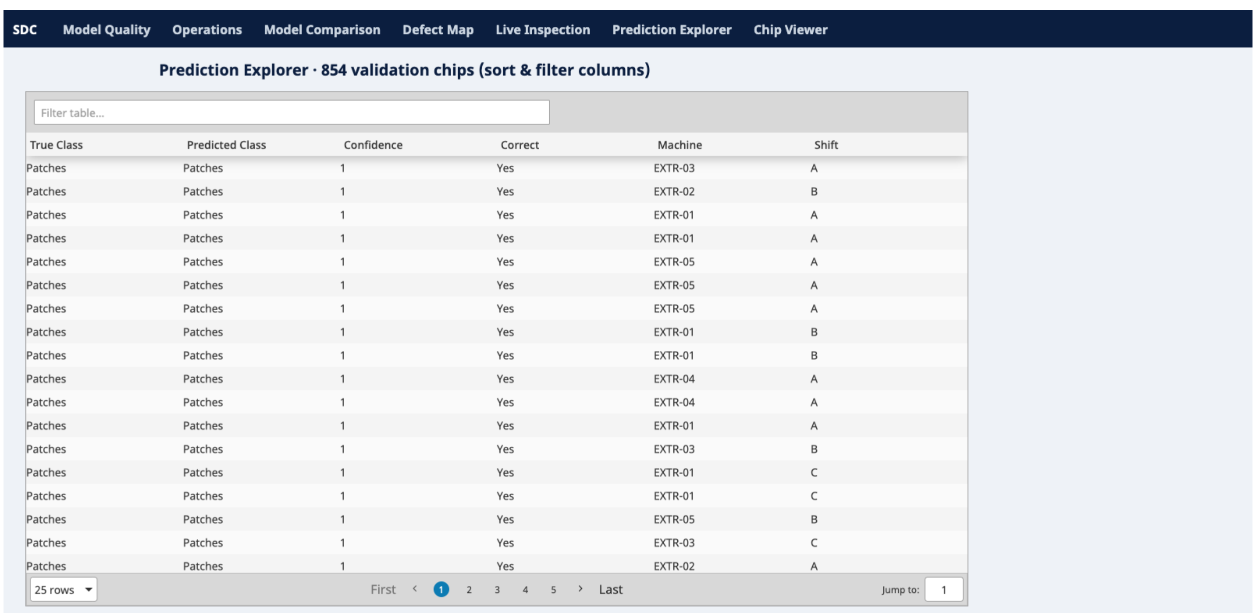
> Want to run it yourself? The dashboard ships as importable Ignition artifacts (a project `.zip`, a full gateway `.gwbk`, and a MySQL dump) under `exports/`. The import steps are in `IMPORT.md`.



Time	Predicted	Actual	Confidence	Severity	Machine	Shift
15:09:41	Patches	Patches	1	CRITICAL	EXTR-04	A
15:09:41	Inclusion	Inclusion	0.99		EXTR-01	C
15:09:40	Pitted Surface	Pitted Surface	1	CRITICAL	EXTR-02	A
15:09:40	Scratches	Scratches	1		EXTR-03	A
15:09:40	Crazing	Crazing	1		EXTR-01	B
15:09:39	Rolled-in Scale	Rolled-in Scale	0.76		EXTR-04	A
15:09:39	Inclusion	Inclusion	1		EXTR-04	C
15:09:39	Rolled-in Scale	Rolled-in Scale	1		EXTR-01	C
15:09:38	Rolled-in Scale	Rolled-in Scale	0.98		EXTR-05	B
15:09:38	Crazing	Crazing	1		EXTR-05	B
15:09:38	Crazing	Rolled-in Scale	0.54		EXTR-01	B
15:09:37	Inclusion	Inclusion	1		EXTR-05	A
15:09:37	Inclusion	Inclusion	1		EXTR-04	B
15:09:37	Patches	Patches	1		EXTR-02	B
15:09:36	Crazing	Crazing	1		EXTR-03	B
15:09:36	Pitted Surface	Pitted Surface	0.82		EXTR-01	C
15:09:36	Patches	Patches	1		EXTR-01	A

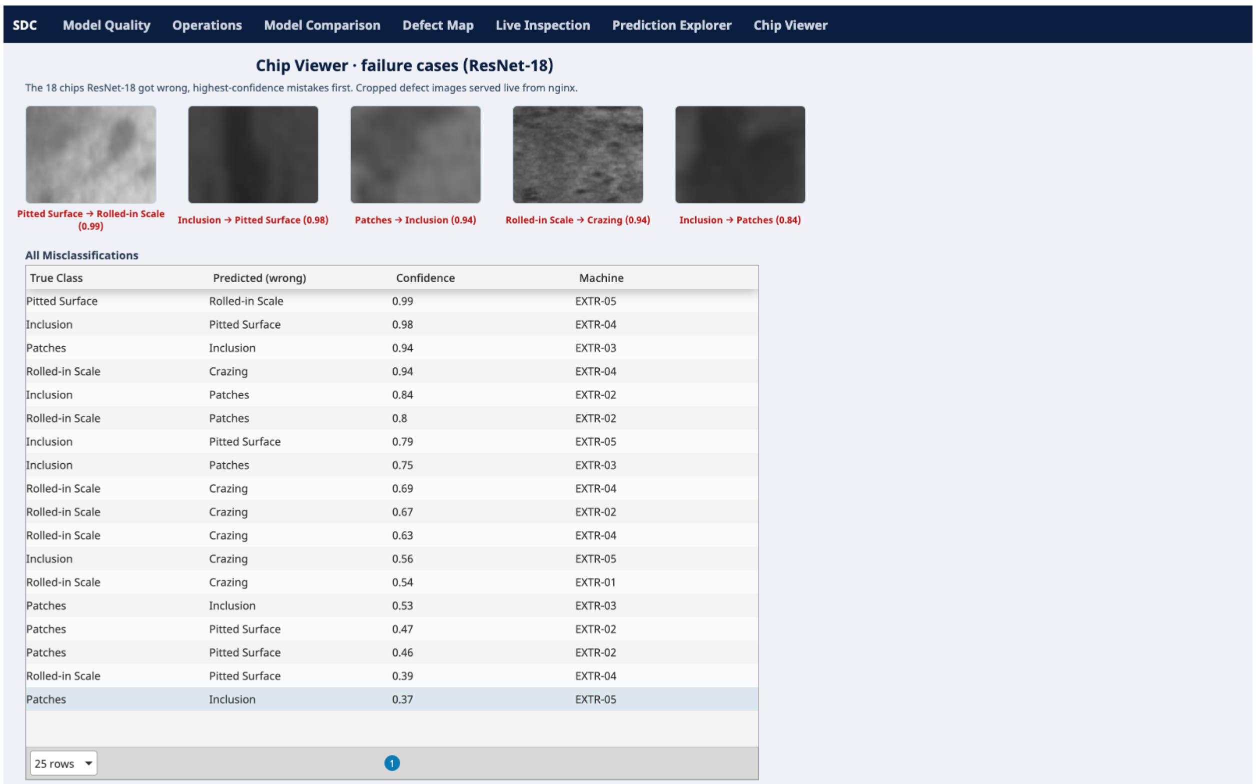
Predicted Class	Count	% Critical
Patches	1,063	14
Crazing	901	14
Inclusion	844	33
Rolled-in Scale	692	32
Scratches	661	17
Pitted Surface	466	39

Figure 5: Live Inspection



True Class	Predicted Class	Confidence	Correct	Machine	Shift
Patches	Patches	1	Yes	EXTR-03	A
Patches	Patches	1	Yes	EXTR-02	B
Patches	Patches	1	Yes	EXTR-01	A
Patches	Patches	1	Yes	EXTR-01	A
Patches	Patches	1	Yes	EXTR-05	A
Patches	Patches	1	Yes	EXTR-05	A
Patches	Patches	1	Yes	EXTR-05	A
Patches	Patches	1	Yes	EXTR-01	B
Patches	Patches	1	Yes	EXTR-01	B
Patches	Patches	1	Yes	EXTR-04	A
Patches	Patches	1	Yes	EXTR-04	A
Patches	Patches	1	Yes	EXTR-01	A
Patches	Patches	1	Yes	EXTR-03	B
Patches	Patches	1	Yes	EXTR-01	C
Patches	Patches	1	Yes	EXTR-01	C
Patches	Patches	1	Yes	EXTR-05	B
Patches	Patches	1	Yes	EXTR-03	C
Patches	Patches	1	Yes	EXTR-02	A

Figure 6: Prediction Explorer



True Class	Predicted (wrong)	Confidence	Machine
Pitted Surface	Rolled-in Scale	0.99	EXTR-05
Inclusion	Pitted Surface	0.98	EXTR-04
Patches	Inclusion	0.94	EXTR-03
Rolled-in Scale	Crazing	0.94	EXTR-04
Inclusion	Patches	0.84	EXTR-02
Rolled-in Scale	Patches	0.8	EXTR-02
Inclusion	Pitted Surface	0.79	EXTR-05
Inclusion	Patches	0.75	EXTR-03
Rolled-in Scale	Crazing	0.69	EXTR-04
Rolled-in Scale	Crazing	0.67	EXTR-02
Rolled-in Scale	Crazing	0.63	EXTR-04
Inclusion	Crazing	0.56	EXTR-05
Rolled-in Scale	Crazing	0.54	EXTR-01
Patches	Inclusion	0.53	EXTR-03
Patches	Pitted Surface	0.47	EXTR-02
Patches	Pitted Surface	0.46	EXTR-02
Rolled-in Scale	Pitted Surface	0.39	EXTR-04
Patches	Inclusion	0.37	EXTR-05

Figure 7: Chip Viewer

Models & Results

I fine-tuned both models on roughly 3,300 training chips and evaluated them on 854 validation chips.

Metric	ResNet-18	ViT-B/16
Accuracy	0.9789	0.7658
Macro F1	0.9785	0.7431
Weighted F1	0.9789	0.7647

ResNet-18 wins clearly. ViT-B/16 is data-hungry and underfits a set this small, which is a legitimate result to report rather than a failure.

ResNet-18 per-class:

Class	Precision	Recall	F1	Support
Crazing	0.964	1.000	0.982	162
Inclusion	0.981	0.969	0.975	159
Patches	0.984	0.974	0.979	193
Pitted Surface	0.945	0.989	0.966	87
Rolled-in Scale	0.992	0.947	0.969	132
Scratches	1.000	1.000	1.000	121

The one remaining weak spot is a small pitted-surface / rolled-in-scale confusion (the two are texturally similar). It shows up as the top mistake in the Chip Viewer above, where a pitted-surface chip gets predicted as rolled-in scale at 0.99 confidence.

Architecture

```
NEU-DET images + VOC XML
           |
           v
       train.py  --model {resnet18, vit_b_16}
           |
  chip extraction + fine-tuning
           |
           v
  results/<model>/  (predictions, confusion, class report)
           |
           v
  ignition_dashboard/etl/load_mysql.py  -->  MySQL star schema
           |                                  (+ synthetic ops: machine/operator/shift/co
```

```
         v
  Ignition Perspective  <-- named queries --  MySQL
  (7 live views)
         ^
         \-- cropped chip PNGs served statically to the Chip Viewer
```

The dashboard ships as exported artifacts rather than a running service. The layout of this folder:

```
ignition_dashboard/
  exports/                 # importable deliverables
    surface_defect_project.zip      # Perspective project (views + named queries)
    surface_defect_gateway.gwbk     # full gateway backup (project + DB connection)
    surfaceDefect.sql               # mysqldump: schema + data
  ignition/projects/...    # version-controlled project source
  mysql/01_schema.sql      # star schema
  etl/                     # load_mysql.py, enrich.py, replay.py
  IMPORT.md                # how to stand it up on your own gateway
```

Training (reproducibility)

The NEU-DET dataset is included in `NEU-DET/`, so training runs without a separate download.

```
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt

python train.py --model resnet18 --rebuild-chips   # build chips + train ResNet-18
python train.py --model vit_b_16                    # train ViT-B/16 (reuses chips)
```

`train.py` picks the device automatically (`cuda`, then `mps`, then `cpu`). Outputs land in `results/<model>/` per model, and the shared chips go to `results/chips/` with index files at `results/*_chips_index.csv`.

The six NEU-DET classes are Crazing, Inclusion, Patches, Pitted Surface, Rolled-in Scale, and Scratches.

Limitations

- ViT-B/16 underperforms on this small dataset. A larger corpus would change the comparison.
- The factory-operations layer (machine, operator, shift, cost, timeline, strip position) in the dashboard is synthetic. I generated it deterministically to demonstrate an MES-style inspection view, and it is not real plant data.

- The benchmark may not represent plant-specific lighting, coating, or imaging conditions.
- Localization is treated as preprocessing, not as a jointly learned detection task. "'